Three-Dimensional Hoek-Brown Strength Criterion for Rocks

Lianyang Zhang[1] and Hehua Zhu[2]

Abstract: A great number of rock strength criteria have been proposed over the past decades. Of these different strength criteria, the Hoek-Brown strength criterion has been used most widely, because: (1) it has been developed specifically for rock materials and rock masses; (2) its input parameters can be determined from routine unconfined compression tests, mineralogical examination, and discontinuity characterization; and (3) it has been applied for over 20 years by practitioners in rock engineering, and has been applied successfully to a wide range of intact and fractured rock types. The Hoek-Brown strength criterion, however, does not take account of the influence of the intermediate principal stress, although much evidence has been accumulating to indicate that the intermediate principal stress does influence the rock strength in many instances. In this paper, a three-dimensional (3D) version of the Hoek-Brown strength criterion has been proposed. The original Hoek-Brown strength criterion is just a two-dimensional (2D) version of the proposed 3D strength criterion. The 3D strength criterion not only inherits the advantages of the original Hoek-Brown strength criterion, but can take account of the influence of the intermediate principal stress. Polyaxial or true triaxial compression test data of intact rocks and jointed rock masses has been collected from the published literature and used to validate the proposed 3D Hoek-Brown strength criterion. Predictions of the proposed 3D Hoek-Brown strength criterion are in good agreement with the test data for a range of different rock types. The proposed 3D Hoek-Brown strength criterion is also compared with a simplified 3D Hoek-Brown strength criterion proposed by Pan and Hudson. The Pan-Hudson criterion cannot be considered a true 3D version of the Hoek-Brown criterion, because it does not reduce to the form of the original Hoek-Brown criterion at either triaxial or biaxial state. The Pan-Hudson criterion underpredicts the strength at the triaxial state, but overpredicts the strength at the biaxial state.

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Introduction

Analysis of a wide range of problems in rock mechanics and rock engineering requires determination of the strength of the rock masses. Different strength criteria have been developed for rocks. Pan and Hudson (1988) provide a good summary of rock strength criteria up to 1988. Colmenares and Zoback (2002) provide a comprehensive description and statistical evaluation of strength criteria that have been most commonly applied to rocks. Of these different rock strength criteria, the Hoek–Brown criterion has been used the most widely, for the following reasons (Priest 2005):

- The Hoek–Brown criterion has been developed specifically for rock materials and rock masses;
- Input parameters for the Hoek–Brown criterion can be derived from uniaxial testing of the rock material, mineralogical examination, and characterization of the rock discontinuities; and
- The Hoek–Brown criterion has been applied for over 20 years by practitioners in rock engineering, and has been applied successfully to a wide range of intact and fractured rock types (Hoek and Brown 1997).

The Hoek–Brown strength criterion is an empirical criterion which was developed by trial and error and is based on the observed behavior of rock masses, model studies to simulate the failure mechanism of jointed rock, and triaxial compression tests of the fractured rock (Hoek and Brown 1980, 1988; Zhang and Einstein 1998). For intact rock, the Hoek–Brown strength criterion can be expressed in the following form

$$\sigma_1' = \sigma_3' + \sigma_c\left(m_i\frac{\sigma_3'}{\sigma_c} + 1\right)^{0.5} \tag{1}$$

where σ_c=unconfined compressive strength of the intact rock; σ_1' and σ_3' are, respectively, the major and minor effective principal stresses; and m_i=material constant for the intact rock. m_i depends only upon the rock type (texture and mineralogy) and can be selected from Table 1 (Hoek and Brown 1997; Marinos and Hoek 2001). For jointed rock masses, the Hoek–Brown strength criterion is given by the equation

$$\sigma_1' = \sigma_3' + \sigma_c\left(m_b\frac{\sigma_3'}{\sigma_c} + s\right)^{0.5} \tag{2}$$

where m_b=material constant for the rock mass; and s=constant that depends on the characteristics of the rock mass. The material constants m_b and s can be estimated from the rock mass rating (RMR) as follows (Hoek and Brown 1988):

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Table 1. Values of Parameter m_i for Different Rocks (Hoek and Brown 1997; Marinos and Hoek 2001)

Rock type	Class	Group	Texture: Coarse	Medium	Fine	Very fine
Sedimentary	Clastic		Conglomerate (21 ± 3)[a] Breccia (19 ± 5)	Sandstone 17 ± 4	Siltstone 7 ± 2 Greywacke (18 ± 3)	Claystone 4 ± 2 Shale (6 ± 2) Marl (7 ± 2)
	NonClastic	Carbonate	Crystalline Limestone (12 ± 3)	Sparitic Limestone (10 ± 2)	Micritic Limestone (9 ± 2)	Dolomite (9 ± 3)
		Evaporite		Gypsum 8 ± 2	Anhydrite 12 ± 2	
		Organic				Chalk 7 ± 2
Metamorphic	Nonfoliated		Marble 9 ± 3	Hornfels (19 ± 4) Metasandstone (19 ± 3)	Quartzite 20 ± 3	
	Slightly foliated		Migmatite (29 ± 3)	Amphibolite 26 ± 6	Gneiss 28 ± 5	
	Foliated[b]			Schist 12 ± 3	Phyllite (7 ± 3)	Slate 7 ± 4
Igneous	Plutonic	Light	Granite 32 ± 3 Granodiorite (29 ± 3)	Diorite 25 ± 5		
		Dark	Gabbro 27 ± 3 Norite 20 ± 5	Dolerite (16 ± 5)		
	Hypabyssal		Porphyrie (20 ± 5)		Diabase (15 ± 5)	Peridotite (25 ± 5)
	Volcanic	Lava		Rhyolite (25 ± 5) Andesite 25 ± 5	Dacite (25 ± 3) Basalt (25 ± 5)	Obsidian (19 ± 3)
		Pyroclastic	Agglomerate (19 ± 3)	Breccia (19 ± 5)	Tuff (13 ± 5)	

[a]Values in parenthesis are estimates.

[b]These values are for intact rock specimen tests normal to bedding or foliation. The value of m_i will be significantly different if failure occurs along a weakness plane.

1. Disturbed rock masses

$$m_b = \exp\left(\frac{\mathrm{RMR}-100}{14}\right)m_i \tag{3a}$$

$$s = \exp\left(\frac{\mathrm{RMR}-100}{6}\right) \tag{3b}$$

2. Undisturbed or interlocking rock masses

$$m_b = \exp\left(\frac{\mathrm{RMR}-100}{28}\right)m_i \tag{4a}$$

$$s = \exp\left(\frac{\mathrm{RMR}-100}{9}\right) \tag{4b}$$

The material constants m_b and s can also be estimated from the updated equations relating m_b and s and the geological strength index (Hoek et al. 1995, 2002; Hoek and Brown 1997).

As can be seen from above, the Hoek–Brown strength criterion does not take account of the influence of the intermediate principal stress. Much evidence, however, has been accumulating to indicate that the intermediate principal stress does influence the rock strength in many instances (Mogi 1971; Pan and Hudson 1988; Brown 1993; Lade 1993; Wang and Kemeny 1995; Chang and Haimson 2000; Colmenares and Zoback 2002; Al-Ajmi and Zimmerman 2005).

In this paper, a three-dimensional (3D) version of the Hoek–Brown strength criterion has been proposed. The proposed 3D Hoek–Brown criterion not only inherits the advantages of the original Hoek–Brown criterion, but can take account of the influence of the intermediate principal stress. Polyaxial or true triaxial compression test data of intact rocks and jointed rock masses has been collected from the published literature and used to validate

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the proposed 3D strength criterion. The difference of the proposed 3D criterion and its advantages over a similar 3D Hoek–Brown strength criterion proposed by Pan and Hudson (1988) are also discussed.

Proposed Three-Dimensional Hoek–Brown Strength Criterion

A 3D Hoek–Brown strength criterion is proposed by modifying the original empirical Hoek–Brown strength criterion so that the influence of the intermediate principal stress can be considered. According to Mogi (1971), a rock strength criterion considering the effect of the intermediate principal stress can be expressed in the following general form (Al-Ajmi and Zimmerman 2005)

$$\tau_{\rm oct} = f(\sigma'_{m,2}) \tag{5}$$

where $\tau_{\rm oct}$ and $\sigma'_{m,2}$ are, respectively, the octahedral shear stress and the effective mean stress defined by

$$\tau_{\rm oct} = \frac{1}{3}\sqrt{(\sigma'_1-\sigma'_2)^2+(\sigma'_2-\sigma'_3)^2+(\sigma'_3-\sigma'_1)^2} \tag{6}$$

$$\sigma'_{m,2} = \frac{\sigma'_1+\sigma'_3}{2} \tag{7}$$

Mogi (1971) suggested the general strength criterion Eq. (5) based on extensive polyaxial compressive tests indicating that the failure plane strikes in the direction of the intermediate principal stress. Al-Ajmi and Zimmerman (2005) validated assumption Eq. (5), by applying it to polyaxial test data for different rocks.

Using the general form of a rock strength criterion Eq. (5), the following 3D Hoek–Brown strength criterion is proposed by modifying the original empirical Hoek–Brown strength criterion

$$\frac{9}{2\sigma_c}\tau_{\rm oct}^2 + \frac{3}{2\sqrt{2}}m\tau_{\rm oct} - m\sigma'_{m,2} = s\sigma_c \tag{8}$$

For $\sigma'_1=\sigma'_2$ or $\sigma'_2=\sigma'_3$, $\tau_{\rm oct}=\sqrt{2}(\sigma'_1-\sigma'_3)/3$ and Eq. (8) is simplified as

$$\sigma'_1 = \sigma'_3 + \sigma_c\left(m\frac{\sigma'_3}{\sigma_c}+s\right)^{0.5} \tag{9}$$

which is just Eq. (1) for intact rock ($m=m_i$ and $s=1$) and Eq. (2) for rock masses ($m=m_b$). So the original Hoek–Brown strength criterion is just a two-dimensional (2D) version of the proposed general 3D strength criterion, or the proposed criterion is a true 3D version of the Hoek–Brown criterion.

Pan and Hudson (1988) proposed a similar 3D Hoek–Brown strength criterion, which is expressed as

$$\frac{3}{\sigma_c}J_2 + \frac{\sqrt{3}}{2}m\sqrt{J_2} - m\frac{I_1}{3} = s\sigma_c \tag{10}$$

where I_1 and J_2 are, respectively, the first stress invariant, and the second deviatoric stress invariant defined by

$$I_1 = \sigma'_1+\sigma'_2+\sigma'_3 \tag{11}$$

$$J_2 = \frac{(\sigma'_1-\sigma'_2)^2+(\sigma'_2-\sigma'_3)^2+(\sigma'_3-\sigma'_1)^2}{6} \tag{12}$$

Eq. (10) can be rewritten as

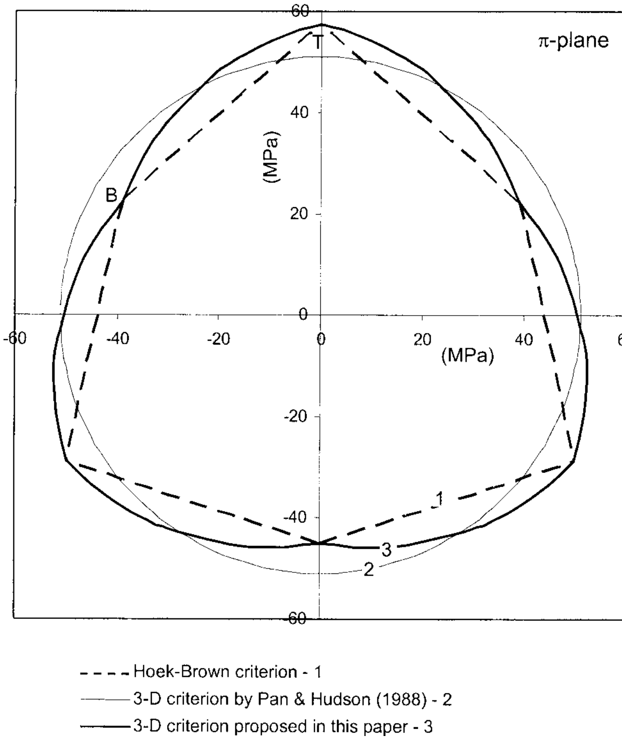

Fig. 1. π-plane plot of Hoek–Brown criterion, 3D criterion by Pan and Hudson (1988) and proposed 3D criterion at $\sigma'_{m,3}=100$ MPa for a rock mass with $\sigma_c=50$ MPa, $m=5$, and $s=0.01$

$$\frac{9}{2\sigma_c}\tau_{\rm oct}^2 + \frac{3}{2\sqrt{2}}m\tau_{\rm oct} - m\sigma'_{m,3} = s\sigma_c \tag{13}$$

in which $\sigma'_{m,3}=I_1/3$ is the effective mean stress.

The only difference between the proposed criterion [Eq. (8)] and the Pan–Hudson criterion [Eq. (13)] is the effective mean stress term. For $\sigma'_1=\sigma'_2$ or $\sigma'_2=\sigma'_3$, the Pan–Hudson criterion will not reduce to the form of the original Hoek–Brown criterion. So the Pan–Hudson criterion cannot be considered a true 3D version of the Hoek–Brown criterion. Fig. 1 shows the π-plane plot of the three criteria at $\sigma'_{m,3}=100$ MPa for a rock mass with $\sigma_c=50$ MPa, $m=5$, and $s=0.01$. The proposed criterion predicts the same strength as the original Hoek–Brown criterion at both triaxial state $\sigma'_2=\sigma'_3$ (Point T) and biaxial state $\sigma'_1=\sigma'_2$ (Point B); but the Pan–Hudson criterion underpredicts the strength at the triaxial state and overpredicts the strength at the biaxial state. Table 2 shows the ratio of $\tau_{\rm oct}$ (Pan–Hudson) to $\tau_{\rm oct}$ (Hoek–Brown) at $\sigma'_{m,3}=100$ MPa for three different rock masses. The difference between $\tau_{\rm oct}$ (Pan–Hudson) and $\tau_{\rm oct}$ (Hoek–Brown) increases as the rock mass becomes harder. Since the Hoek–Brown criterion has been applied successfully to a wide range of intact and fractured rock types, it is essential for a 3D criterion to predict the same strength as the Hoek-Brown criterion at both the triaxial and the biaxial states. So the proposed criterion can be applied to all rock masses to which the original Hoek-Brown criterion applies; but the Pan–Hudson criterion only applies approximately to weak rock mass (i.e., rock mass with small σ_c and/or small m and s).

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Table 2. Ratio of τ_{oct}(Pan–Hudson) to τ_{oct}(Hoek–Brown) at $\sigma'_{m,3}=100$ MPa for Three Different Rock Masses

Rock mass[a]	σ_c (MPa)	m_i	GSI	τ_{oct}(Pan–Hudson)/τ_{oct}(Hoek–Brown) Triaxial state	Biaxial state
Very good quality	150	25	75	0.78	1.24
Average	80	12	50	0.90	1.10
Very poor quality	20	8	30	0.97	1.03

[a]Rock mass quality based on Hoek (2000).

Validation of the Proposed Criterion

In this section, the proposed 3D Hoek–Brown strength criterion is validated by applying it to both intact rocks and jointed rock masses.

Application to Intact Rocks

By searching the literature, the polyaxial or true triaxial compression test data of nine rock types are located. These rock types are KTB Amphibolite, Westerly granite, Dunham dolomite, Apache Leap tuff, Mizuho trachyte, Solenhofen limestone, coarse grained dense marble, Shirahama sandstone, and Yuubari shale (Mogi 1971; Takahashi and Koide 1989; Wang and Kemeny 1995; Haimson and Chang 2000; Chang and Haimson 2000; Colmenares and Zoback 2002; Al-Ajmi and Zimmerman 2005). Since the application of the proposed 3D Hoek–Brown strength criterion requires known value of the unconfined compressive strength σ_c, only the first five rock types with measured σ_c data are selected. The test types for these five rocks are uniaxial, triaxial, and true triaxial.

For KTB Amphibolite, the measured unconfined compressive strength σ_c is 165.0 MPa. According to Table 1 (Hoek and Brown 1997; Marinos and Hoek 2001), $m_i=26\pm6$ for amphibolite. The predictions of the proposed 3D strength criterion, respectively, at the lower bound and upper bound values of $m_i=20$ and 32 are shown in Fig. 2(a). Essentially, all of the test data points fall inside the band formed by the lower bound and upper bound curves. Using the method of least squares (LS), the value of m_i corresponding to the least error (or best fitting) can be obtained as $m_i=31$, which is in the range of $m_i=20-32$ for amphibolite. The error function for the LS method with n test data points is

$$\text{Error} = \sum_{i=1}^{n} (\tau_{\text{oct,experimental}} - \tau_{\text{oct,predicted}})_i^2 \tag{14}$$

For the other four rocks, the lower bound and upper bound values of m_i can be obtained from Table 1, respectively, for Westerly granite, Dunham dolomite, and Apache Leap tuff. Since trachyte is not listed in Table 1, the m_i of Mizuho trachyte is estimated as 13 by best fitting the Hoek–Brown criterion to the triaxial test data. To consider a range, the median of the deviation values 2, 3, 4, 5, and 6 for different rocks in Table 1 is used. So the estimated range of m_i for Mizuho trachyte is $m_i=13\pm4$. The predicted lower bound and upper bound curves for the four rocks are shown, respectively, in Figs. 2(b–e). It can be seen that the test data points either fall inside the band formed by the bound curves or are very close to the bound curves. Using the LS method, the values of m_i corresponding to the best fitting are also obtained for the four rocks, which are all in the range of m_i for the specific rock (see Table 3 and Fig. 2).

Also plotted in Fig. 2 are the test data and the predictions from the Pan–Hudson criterion using the measured σ_c and the lower bound and upper bound values of m_i for each rock. The Pan–Hudson criterion underpredicts the strength for all rocks with the exception only for Mizuho trachyte (which has the lowest σ_c of the five rocks) at the high mean effective stress level. Since most of the test data points are out of the band formed by the bound curves, no best fitting analyses are conducted.

Application to Jointed Rock Masses

The proposed 3D Hoek-Brown strength criterion can also be applied to isotropic jointed rock masses as the original Hoek–Brown strength criterion. Polyaxial or true triaxial compression tests of jointed rock masses have been conducted by a number of investigators including Lögters and Voort (1974), Reik and Zacas (1978), Müller-Salzburg and Ge (1983), Singh et al. (1998), and Tiwari and Rao (2004). Most of these tests, however, have been conducted on anisotropic jointed specimens to study the influence of the intermediate principal stress on the strength of the anisotropic jointed rock masses. In the following, the proposed 3D strength criterion is applied to analyze the jointed biotite granite tested at the Kurobe IV arch dam site in Japan (Müller-Salzburg and Ge 1983; Brown 1993).

Fig. 3(a) shows the test block Q_2 which was 2.8 m long in the σ'_1 direction, 2.8 m high in the σ'_2 direction, and 1.4 m wide in the σ'_3 direction, giving it a total volume of about 11 m^3. The jointed biotite granite was described as "somewhat decomposed," and the unconfined compressive strength of the cores of granite taken from the test location was 23.0 MPa. The test was conducted by increasing and cycling σ'_1 with σ'_2 and σ'_3 held constant at 0.70 and 0.12 MPa, respectively. Fig. 3(b) shows the plot of deformation measured in the σ'_1 direction as σ'_1 was increased and cycled (for the numbers of alternate load cycles, AL, indicated in the figure). From these stress-deformation curves, an initial yield limit was observed at a low stress beyond which considerable irreversible deformation developed. A principal yield limit and a peak or failure stress were also identified. The failure stress was 13.0 MPa, which is much lower than the unconfined compressive strength of the corresponding intact rock.

According to Table 1 (Hoek and Brown 1997; Marinos and Hoek 2001), $m_i=29-35$ for granite. For simplicity, an average value of $m_i=32$ is selected for the analysis. Since the value of RMR is unknown for the tested rock mass, a series of values of RMR is assumed for analyses to create a plot of the failure major principal stress σ'_1 versus RMR. Assuming that the tested rock mass is undisturbed, m_b and s can be calculated from Eqs. (4*a*) and (4*b*) for an assumed RMR, and the corresponding 3D failure major principal stress σ'_1 can be determined from Eq. (8). Fig. 4 shows the variation of the failure major principal stress σ'_1 with RMR from the proposed criterion. For comparison, the failure major principal stress σ'_1 calculated based on the original Hoek–Brown strength criterion [Eq. (2)] and the Pan–Hudson criterion [Eq. (10)] are also shown in the figure. The Pan-Hudson criterion

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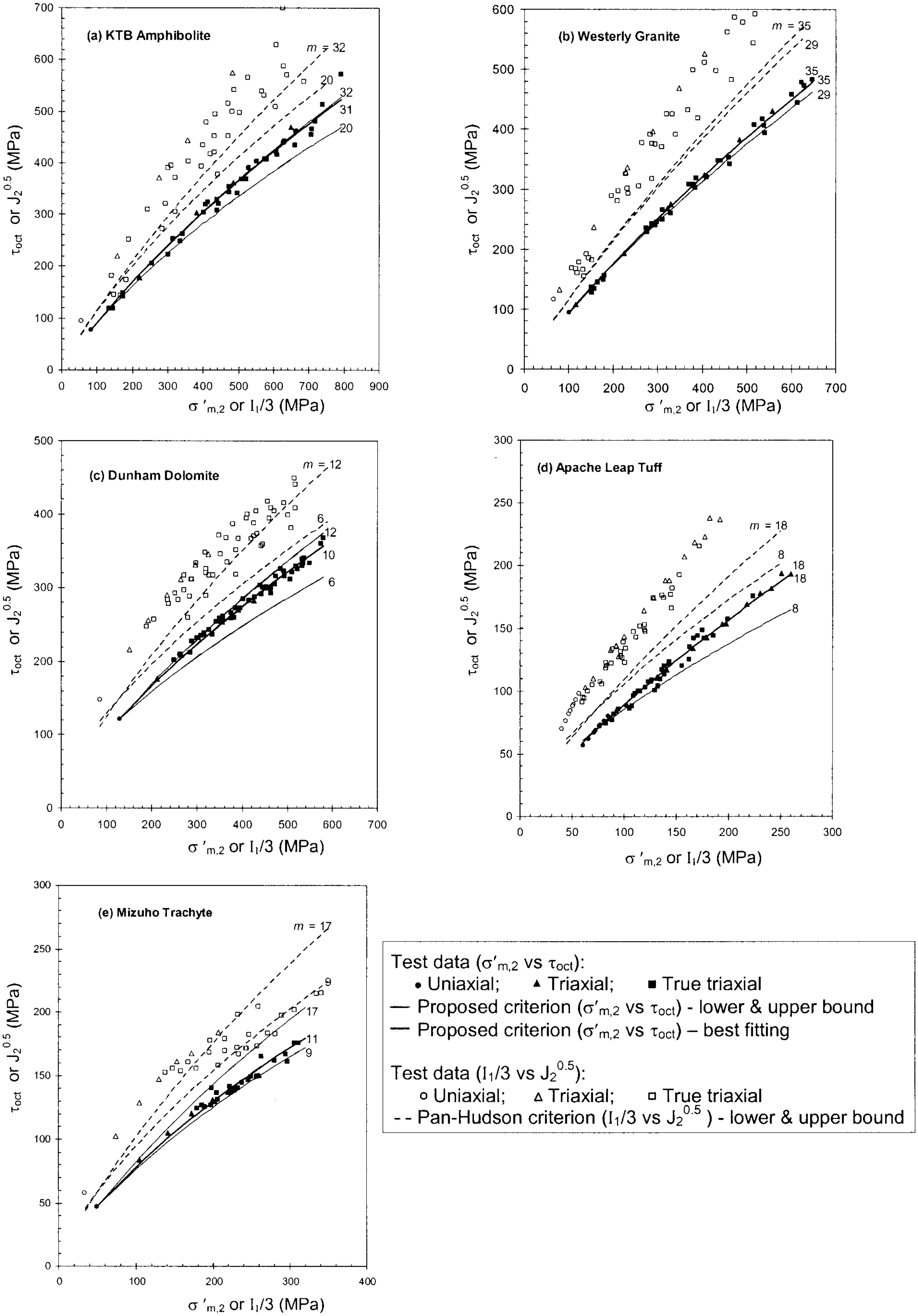

Fig. 2. Comparison of proposed criterion and Pan–Hudson criterion with test data of five intact rocks

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Table 3. Values of σ_c, m_i, and s Used for Predicting Three-Dimensional Strength of Five Intact Rocks

Rock	σ_c (MPa)	m_i Range	m_i Best fitting	s
KTB amphibolite	165.0	26±6[a]	31	1
Westerly granite	201.0	32±3[a]	35	1
Dunham dolomite	257.0	9±3[a]	10	1
Apache leap tuff	147.3	13±5[a]	18	1
Mizuho trachyte	100.0	13±4[b]	11	1

[a]From Table 1.

[b]13 is the estimated value by best-fitting the original Hoek–Brown criterion to the triaxial test data and 4 is simply selected to be the medium of the deviation values 2, 3, 4, 5, and 6 in Table 1.

significantly underpredicts the failure major principal stress σ_1'. The failure major principal stress σ_1' from the proposed criterion is higher than the failure major principal stress σ_1' from the original Hoek–Brown strength criterion, clearly showing the influence of the intermediate principal stress. The measured failure stress σ_1' of 13.0 MPa corresponds to a RMR of about 80 (good quality rock mass) and a RMR of about 86 (very good quality rock mass), respectively, for the proposed 3D strength criterion and the original Hoek–Brown strength criterion. Since the jointed biotite granite was described as somewhat decomposed for the test block Q_2, the tested rock mass should be more likely in good quality than in very good quality. So the proposed 3D strength criterion can better predict the strength of the polyaxially tested jointed rock mass than the original Hoek–Brown strength criterion. Considering the

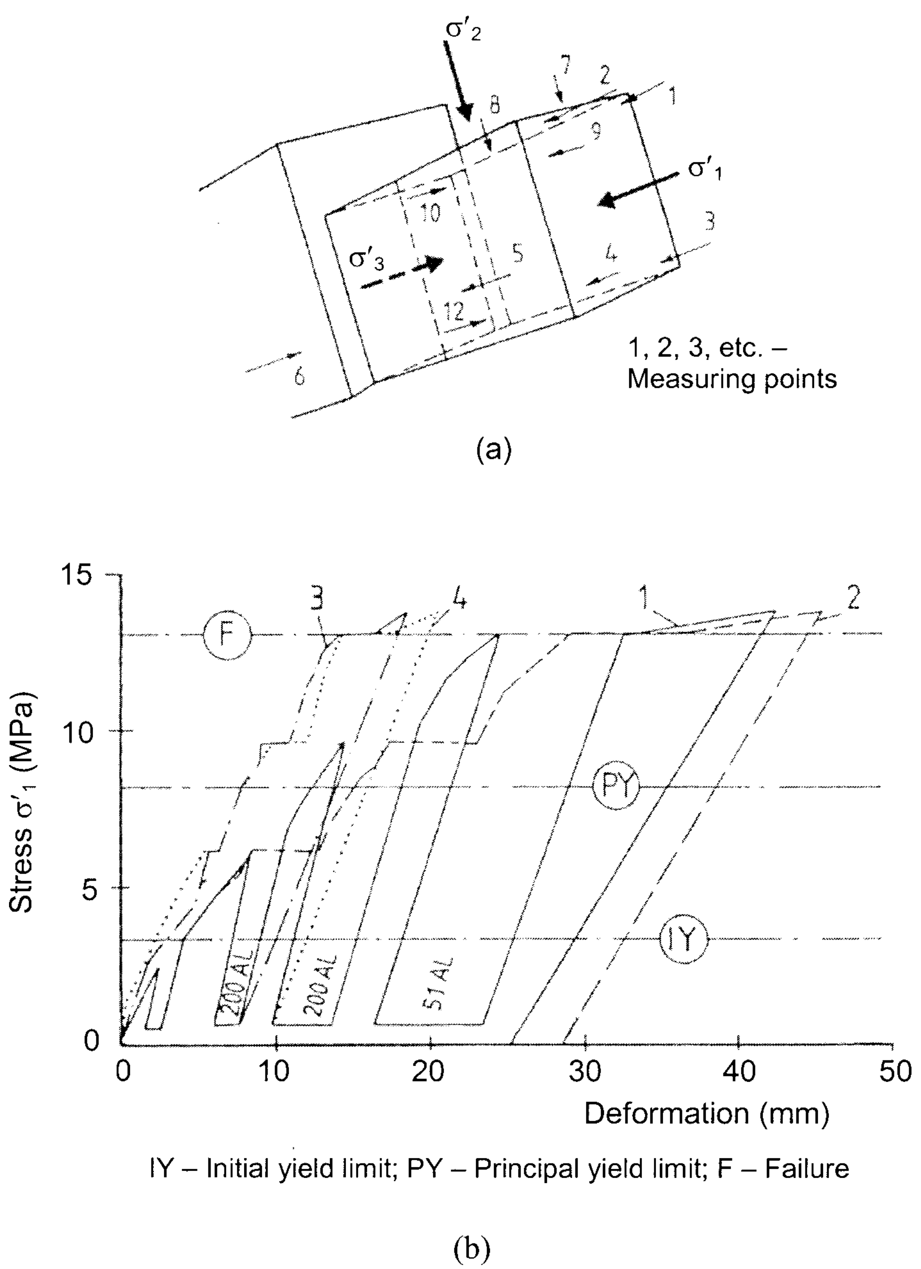

Fig. 3. Polyaxial compression test, Kurobe IV arch dam site, Japan: (a) test block Q_2; (b) major principal compressive stress-deformation curves (Müller-Salzburg and Ge 1983)

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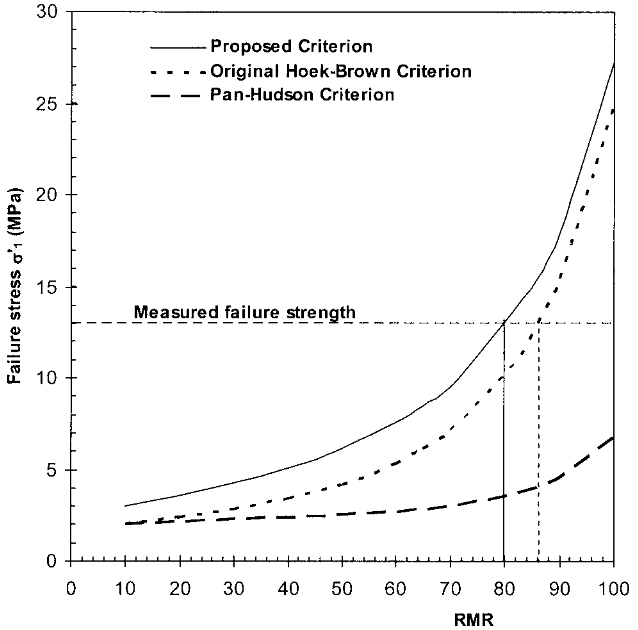

Fig. 4. Variation of failure major principal stress with RMR, respectively, for the proposed criterion, the original Hoek–Brown criterion, and the Pan–Hudson criterion

uncertainty of the RMR value for this test, future true triaxial tests of rock masses are recommended for validating the proposed criterion.

Summary and Conclusions

- Of the many different rock strength criteria, the Hoek–Brown strength criterion has been used most widely, because of its different advantages. The Hoek–Brown strength criterion, however, does not take account of the influence of the intermediate principal stress, although much evidence has been accumulating to indicate that the intermediate principal stress does influence the rock strength.
- A 3D version of the Hoek–Brown strength criterion has been proposed. The original Hoek–Brown strength criterion is just a 2D version of the proposed 3D strength criterion. The proposed 3D strength criterion not only inherits the advantages of the original Hoek–Brown strength criterion, but can take account of the influence of the intermediate principal stress.
- The proposed 3D Hoek–Brown strength criterion has been validated by applying it to polyaxial or true triaxial compression test data of intact rocks and jointed rock masses collected from the published literature.
- The simplified 3D Hoek-Brown strength criterion proposed by Pan and Hudson (1988) cannot be considered a true 3D version of the Hoek–Brown criterion, because it does not reduce to the form of the original Hoek–Brown criterion at either the triaxial or biaxial state. Compared with the Hoek–Brown criterion, the Pan–Hudson criterion underpredicts the strength at the triaxial state, but overpredicts the strength at the biaxial state.
- The proposed 3D Hoek–Brown strength criterion, has the advantage over the existing 3D non-Hoek–Brown type criteria, because it uses the same input parameter as the most widely used Hoek–Brown criterion.

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